

HAONAN GE

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Ph.D. Student in Computer Science GitHub: [Haonan-Ge](#)
Google Scholar

RESEARCH INTERESTS Large Language Model, Video Understanding, Agentic AI, World Model.

EDUCATION **University of California, Santa Barbara**, Santa Barbara, CA Sep 2026 - Present
Ph.D. in Computer Science
Advisor: Prof. [Yao Qin](#)
Southeast University, Nanjing, China Sep 2022 - Jun 2026
B.Eng. in Information Engineering
GPA: 3.72/4.0.
University of California, Irvine, Irvine, CA Jun 2025 - Sep 2025
Exchange in Computer Science
GPA: 4.0/4.0.

RESEARCH POSITIONS **University of California, Los Angeles & The University of Queensland** May 2025 - Jun 2026
Research Intern (Remote) (Advisors: Prof. [Yujun Cai](#) and Prof. [Kai-Wei Chang](#))
University of California, Irvine & Rice University Jun 2025 - Sep 2025
Research Intern (On-site) (Advisors: Prof. [Weining Shen](#) and Prof. [Hanjie Chen](#))
The University of Queensland Feb 2025 - May 2025
Research Intern (Remote) (Advisor: Prof. [Yujun Cai](#))

PUBLICATIONS [Google Scholar](#)

UNDER REVIEW AND PREPRINTS

- [3] **Haonan Ge**, Yiwei Wang, Kai-Wei Chang, Hang Wu, and Yujun Cai. “FrameMind: Frame-Interleaved Video Reasoning via Reinforcement Learning.” *Under review at NeurIPS 2026*. [\[PDF\]](#)
- [2] Hang Wu, Yujun Cai, Zehao Li, **Haonan Ge**, Bowen Sun, Junsong Yuan, and Yiwei Wang. “CamReasoner: Reinforcing Camera Movement Understanding via Structured Spatial Reasoning.” *Under review at NeurIPS 2026*. [\[PDF\]](#)
- [1] Hang Wu, Yujun Cai, **Haonan Ge**, Hongkai Chen, Ming-Hsuan Yang, and Yiwei Wang. “RefineShot: Rethinking Cinematography Understanding with Foundational Skill Evaluation.” *arXiv preprint arXiv:2510.02423, 2025*. [\[PDF\]](#)

PEER-REVIEWED PUBLICATIONS

- [4] **Haonan Ge**, Yiwei Wang, Hang Wu, and Yujun Cai. “What Should a Streaming Video Model Remember?” *NeurIPS 2026*. [\[PDF\]](#)
[Project website](#).
- [3] Yixu Huang, Bo Li, Na Li, Zhe Wang, Kaijie Chen, **Haonan Ge**, Qingyi Si, Yuanzhe Shen, Ruihan Yang, Guangjing Wang, and Hongcheng Guo. “GUI Agents for Continual Game Generation.” *EMNLP 2026*. [\[PDF\]](#)
[Project website](#).

- [2] Haotian Xia*, **Haonan Ge***, Junbo Zou*, Hyun Woo Choi, Xuebin Zhang, Danny Suradja, Botao Rui, Ethan Tran, Wendy Jin, Zhen Ye, Xiyang Lin, Christopher Lai, Shengjie Zhang, Junwen Miao, Shichao Chen, Rhys Tracy, Vicente Ordonez, Weining Shen, and Hanjie Chen. “SportR: A Benchmark for Multimodal Large Language Model Reasoning in Sports.” *ICLR 2026*. [\[PDF\]](#)
[Code](#). * Equal contribution.
- [1] **Haonan Ge**, Yiwei Wang, Ming-Hsuan Yang, and Yujun Cai. “MRFD: Multi-Region Fusion Decoding with Self-Consistency for Mitigating Hallucinations in LVLMS.” *Findings of EMNLP 2025*. [\[PDF\]](#)
[Project website](#).

RESEARCH EXPERIENCE

Selective Memory for Streaming Video Understanding (Lead) May 2025 - Present
University of California, Los Angeles & The University of Queensland
Paper: “What Should a Streaming Video Model Remember?” *NeurIPS 2026*.

- Proposed **SelectStream**, casting streaming memory as **budgeted online latent evidence allocation** over a fixed-capacity latent-memory graph with surprise-driven windowing and priority-preserving consolidation.
- Added query-conditioned graph reasoning with calibrated latent evidence injection (no frame replay or context growth), reaching **82.67%** on StreamingBench, **67.03%** on OVO-Bench, and **74.4%** on offline benchmarks with near-flat latency and GPU memory.

Frame-Interleaved Video Reasoning via Reinforcement Learning (Lead) May 2025 - Present
University of California, Los Angeles & The University of Queensland
Paper: “FrameMind: Frame-Interleaved Video Reasoning via Reinforcement Learning.” *Under review at NeurIPS 2026*.

- Proposed **Frame-Interleaved CoT (FiCOT)**, a multi-turn loop in which the model alternates textual reasoning with *requesting* targeted frames or clips as visual evidence.
- Introduced **Dynamic-Resolution Frame Sampling (DRFS)** and **DRFS-GRPO** to learn temporal-spatial sampling policies from outcome rewards, without frame-level labels.
- Built a compact **agent runtime** (tool schema, parsing/verification, 1–3 turn cap); achieved strong gains on long-video benchmarks (MLVU, VideoMME) with better efficiency and interpretability.

Camera Movement Understanding via Structured Spatial Reasoning May 2025 - Present
University of California, Los Angeles & The University of Queensland
Paper: “CamReasoner: Reinforcing Camera Movement Understanding via Structured Spatial Reasoning.” *Under review at NeurIPS 2026*.

- Co-developed the **Observation-Thinking-Answer (O-T-A)** paradigm, reformulating camera movement recognition as structured ego-motion inference; built **18k SFT reasoning chains** and **38k RL samples**.
- Trained with two-stage SFT and **EMA-GRPO**, improving Qwen2.5-VL-7B from 73.8% to **78.4%** on binary classification and from 60.9% to **74.5%** on VQA.

Multimodal LLM Reasoning Benchmark for Sports (Co-lead) Jun 2025 - Sep 2025
University of California, Irvine & Rice University
Paper: “SportR: A Benchmark for Multimodal Large Language Model Reasoning in Sports.” *ICLR 2026*.

- Led **badminton** CoT annotation and quality control; designed experiments for infraction type, penalty, explanation, tactics, and **grounding** tasks.
- Implemented the **SFT** and **GRPO**-based **RL** pipeline (LLM-as-Judge rewards for QA, **GIoU** for grounding); ablations validated SFT → RL gains and cross-modal generalization.

- Benchmarked open-source MLLMs (Qwen-VL, LLaVA-Next, GLM-4.5V, DeepSeek-VL) with unified prompts and a judge ensemble.

Hallucination Mitigation in LVLMS (First Author)

Feb 2025 - May 2025

The University of Queensland

Paper: “MRFD: Multi-Region Fusion Decoding with Self-Consistency for Mitigating Hallucinations in LVLMS.” *Findings of EMNLP 2025*.

- Proposed a **training-free** decoding method that selects salient regions via cross-attention, queries each with **region-aware prompts**, and fuses per-region responses using **Jensen–Shannon Divergence**–based reliability weights.
- Reduced hallucinations without modifying model parameters, with consistent gains over strong decoding baselines on POPE, CHAIR, and MME-Hallucination.

**HONORS AND
AWARDS**

Outstanding Student Award

2025

Southeast University, Nanjing, China.

**PROFESSIONAL
SERVICES**

Reviewer/Program Committee

- ICLR (2027), NeurIPS (2026), CVPR (2026), WACV (2026, 2027)
- ACM MM (2026), CIKM (2026), ACL Rolling Review (2026)

SKILLS

Programming: Python, C/C++, Matlab, HTML/CSS/JS, $\text{\LaTeX}$.

ML/Systems: PyTorch, Ray, FSDP, NumPy/Pandas, Git, Docker.

OS: Linux (Ubuntu), Windows.

Languages: Mandarin (Native), English (Fluent).